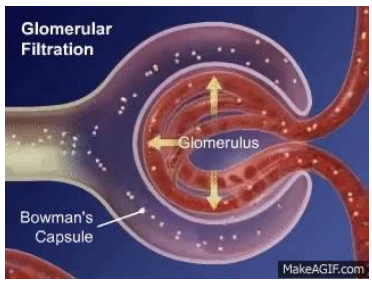
Physiology of Urine Formation

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Contents:

- Introduction.
- Processes of Urine Formation
- Glomerular Filtration Rate (GFR)
- Pressures Determining Filtration
- Filtration Fraction and Filtration Coefficient
- The Site and Routes of Reabsorption

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Objectives:

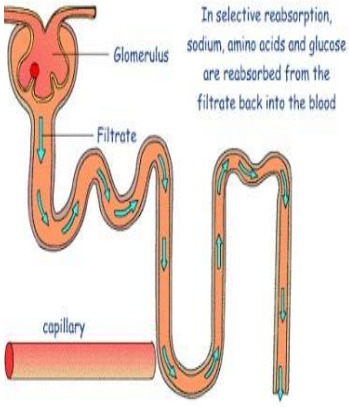
By the end of this lecture, you will be able to:

- Describe the Processes of Urine Formation
- Define the Glomerular Filtration Rate (GFR)
- Know the Pressures that are Determining Filtration
- Filtration Fraction and Filtration Coefficient
- Know the Site and Routes of Reabsorption

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Introduction:

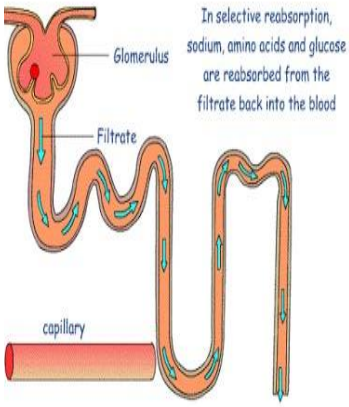
- The kidney glomerulus filters blood mainly based on particle size to produce a filtrate lacking cells or large proteins.
- Most of the ions and molecules in the filtrate are needed by the body and must be reabsorbed farther down the nephron tubules, resulting in the formation of urine.



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Introduction:

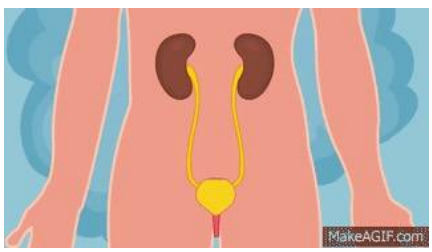
- Many substances that need to be removed from the body still remain in the blood.
- The tubule cells remove them from the blood and secrete them into the filtrate, thereby removing them from the body.



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Introduction:

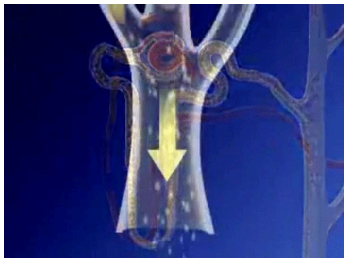
- Urine formation is the **blood-cleansing** function.
- Normally, **~1300 mL** of blood enters the kidneys (=26% of cardiac output).

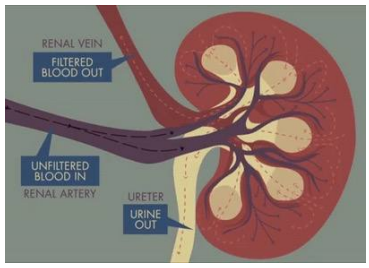


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Introduction:

- The kidneys **excrete the unwanted substances** along with water from the blood as **urine**.
- Normal urinary output is **1 - 1.5 L/day**.

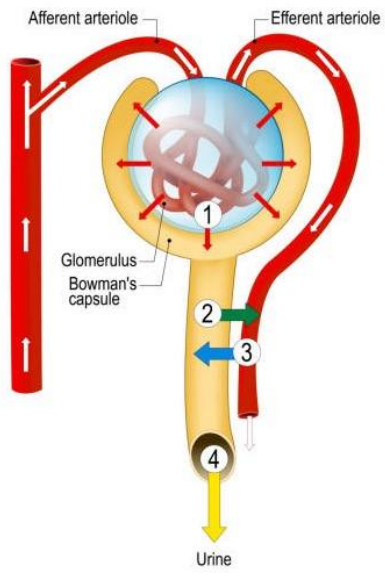




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Processes of Urine Formation:

1. **Glomerular filtration:**
2. **Tubular reabsorption:**
3. **Tubular secretion:**

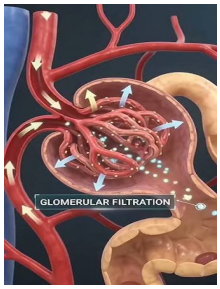


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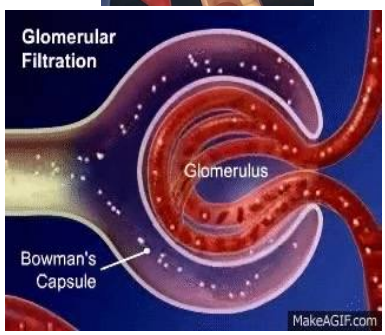
Processes of Urine Formation:

1. Glomerular filtration:

- Blood passes through glomerular capillaries, the plasma is filtered into the Bowman's capsule.



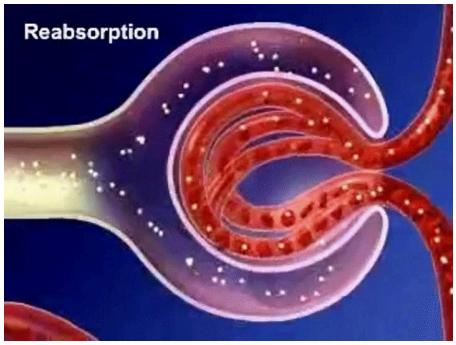
- The **filtrate** passing through the tubule, undergoes various changes both in **quality/quantity**.



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Processes of Urine Formation:

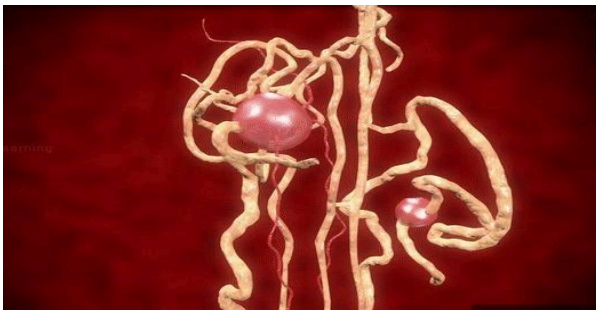
- Many **wanted substances** like glucose, amino acids, water and electrolytes are **reabsorbed** from the tubules (tubular reabsorption).



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Processes of Urine Formation:

- some **unwanted substances** are secreted into the tubule from peritubular blood vessels (tubular secretion or excretion).

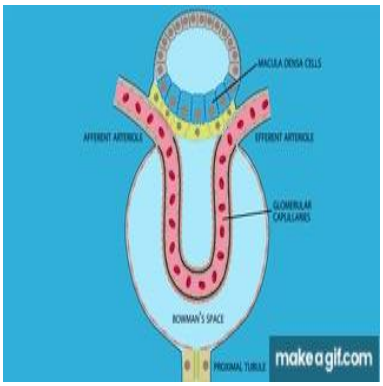


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Processes of Urine Formation:

Ultrafiltration:

- **Glomerular filtration** is called **ultrafiltration** because even the minute particles are filtered. But the **plasma proteins are not filtered** due to their large molecular size.
- The protein molecules are larger than the filtration pores present in the **endothelium of capillaries**.



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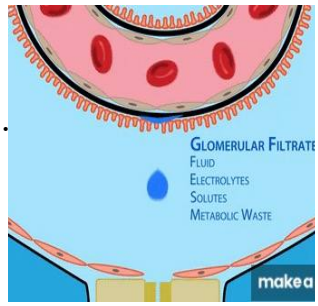
Processes of Urine Formation:

Glomerular Filtration Rate (GFR):

- GFR is defined as **total quantity of filtrate** formed in all nephrons of both the kidneys in a given unit of time.

Normal GFR in a healthy adult:

- Normal GFR is **~125 mL/min**.
- **180 L. filtrate/day** by the 2 kidneys.

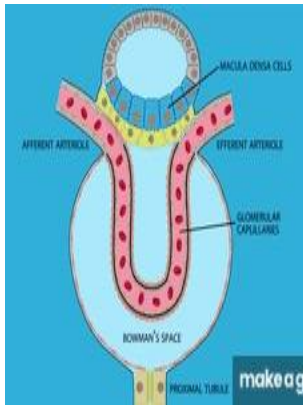


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Processes of Urine Formation:

Filtration Fraction:

- The proportion of blood plasma that enters the kidneys and is successfully filtered across the glomerular capillaries into Bowman's capsule.
- It is the **fraction of the renal plasma, which becomes the filtrate**.
- It is the **ratio between renal plasma flow and GFR**.



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Processes of Urine Formation:

Pressures Determining Filtration:

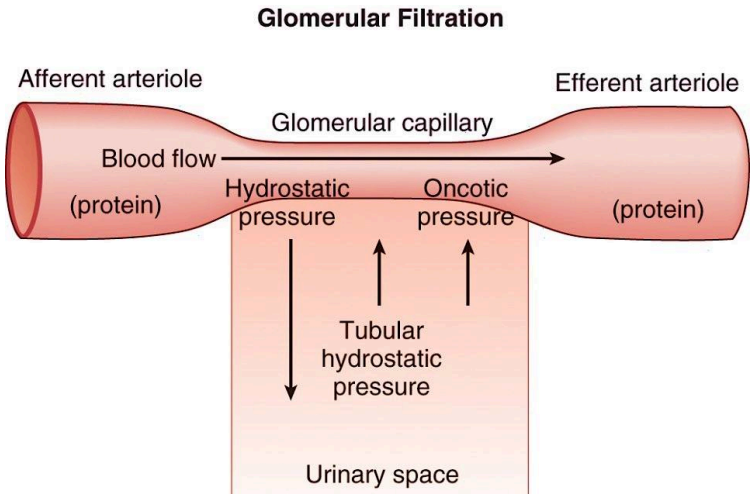
1. **Glomerular capillary pressure**
2. **Colloidal osmotic pressure in the glomeruli**
3. **Hydrostatic pressure in the Bowman's capsule**

- These pressures **determine the GFR by either favoring or opposing the filtration**.

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Processes of Urine Formation:

Pressures Determining Filtration:



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Processes of Urine Formation:

Pressures Determining Filtration:

1. **Glomerular capillary pressure:**
   - The pressure exerted by the **blood** in glomerular capillaries.
   - is **~60 mm Hg** (varies: **45 and 70 mmHg**).
   - is the **highest** capillary pressure in the body.
   - it **favors glomerular filtration**.

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Processes of Urine Formation:

Pressures Determining Filtration:

2. **Colloidal osmotic pressure in the glomeruli**
   - exerted by **plasma proteins** in the glomeruli.
   - The plasma proteins are **not filtered** through the glomerular capillaries and remain in the glomerular capillaries.
   - It is **~25 mm Hg**.
   - It **opposes glomerular filtration**.

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Processes of Urine Formation:

Pressures Determining Filtration:

3. Hydrostatic pressure in the Bowman's capsule

- exerted by the **filtrate** **in Bowman's capsule**.
- It is also called **capsular pressure**.
- It is **~15 mm Hg.**
- It also **opposes glomerular filtration**.

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Processes of Urine Formation:

Net Filtration Pressures (NFP):

- **is the balance between pressure favoring filtration and pressures opposing filtration.**
- It is otherwise known as **effective (essential) filtration pressure**.

Net Filtration Pressures = Glomerular capillary pressure − {Colloidal osmotic pressure + Hydrostatic pressure in Bowman's capsule}

= 60 – (25 + 15) = 20 mm Hg

- **NFP is ~20 mm Hg** (varies: **15 and 20 mm Hg**)

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Processes of Urine Formation:

Filtration Coefficient (FC):

- Is the GFR in terms of net filtration pressure.
- It is the GFR per mmHg of NFP.
- For example, when GFR is 125 mL/min and NFP is 20 mm Hg:

- **Filtration Coefficient** $= \frac{125\ \text{mL}}{20\ \text{mm Hg}}$

$= 6.25\ \text{mL/mm Hg}$

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Factors Affecting GFR:

- Renal Blood Flow
- Tubuloglomerular Feedback
- Glomerular Capillary Pressure
- Colloidal Osmotic Pressure
- Pressure in Bowman's Capsule
- Constriction of Afferent Arteriole
- Constriction of Efferent Arteriole

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Factors Affecting GFR: cont….

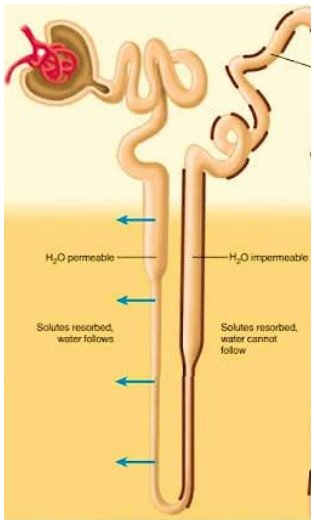
- Systemic Arterial Pressure
- Sympathetic Stimulation
- Surface Area of Capillary Membrane
- Permeability of Capillary Membrane
- Hormonal and Other Factors

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Processes of Urine Formation:

2. Tubular reabsorption:

- The process by which water electrolytes and other **substances are transported from renal tubules back to the blood**.
- When the glomerular filtrate flows through the tubular portion of nephron, both **quantitative** and **qualitative** changes occur.



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Processes of Urine Formation:

2. Tubular reabsorption:

- Large quantity of water (>99%), electrolytes and other substances are reabsorbed by the tubular epithelial cells.
- The reabsorbed substances move into the interstitial fluid of renal medulla and then move into the blood in peritubular capillaries.

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Processes of Urine Formation:

2. Tubular reabsorption:

- It is known as **selective reabsorption** because the tubular cells reabsorb only the substances necessary for the body.
  - **Essential substances** (glucose and amino acids) are completely reabsorbed from renal tubule.
  - **Unwanted substances** (metabolic waste products) are not reabsorbed and excreted through urine.

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Processes of Urine Formation:

Routes of Reabsorption:

- Reabsorption of substances from tubular lumen into the peritubular capillary occurs by 2 routes:

1. Transcellular route

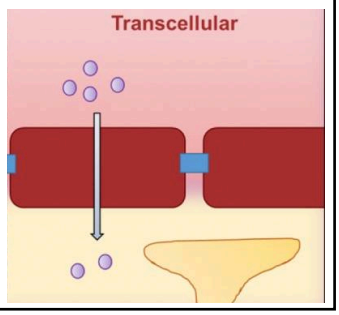
2. Paracellular route

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Processes of Urine Formation:

Transcellular route:

- Substances move through the cell.
- **It includes transport of substances from:**
  - Tubular lumen into tubular cell through apical (luminal) surface of the cell membrane
  - Tubular cell into interstitial fluid.
  - Interstitial fluid into capillary

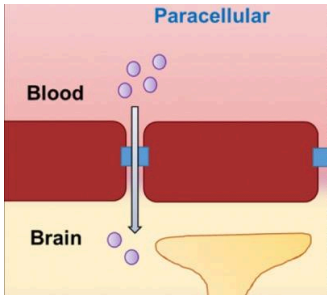


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Processes of Urine Formation:

Paracellular route:

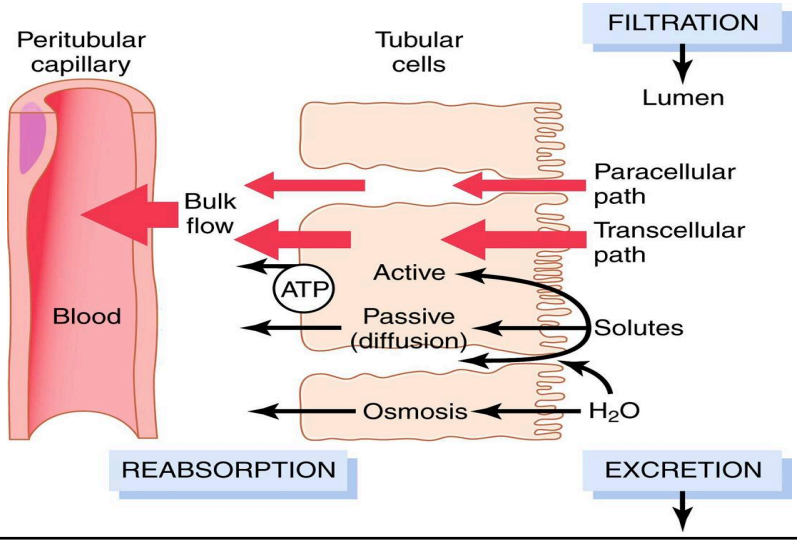
- Substances move through the intercellular space.
- **It includes transport of substances from:**
  - Tubular lumen into interstitial fluid present in lateral intercellular space through the tight junction between the cells.
  - Interstitial fluid into capillary



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Processes of Urine Formation:

Routes of Reabsorption:



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Processes of Urine Formation:

Site of Reabsorption:

- Reabsorption of the substances occurs in almost all the segments of tubular portion of nephron.

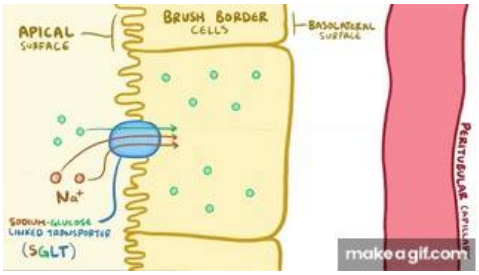
- **Substances Reabsorbed from:**
  - ✓ **Proximal Convoluted Tubule (PCT)**
  - ✓ **Loop of Henle**
  - ✓ **Distal Convoluted Tubule (DCT)**
  - ✓ **Collecting Duct (CD)**

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Processes of Urine Formation:

Tubular transport maximum (Tm):

- is the rate at which the **maximum amount** of a substance is reabsorbed from the renal tubule.
- for every actively reabsorbed substance, there is a maximum rate at which it could be reabsorbed.



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Processes of Urine Formation:

Reabsorption of Important Substance:

- **Reabsorption of Sodium**
  - 99% of sodium is reabsorbed.
  - 2/3 of Na+ (66%) is reabsorbed in PCT and CD.
- **Reabsorption of water :**
  - occurs from PCT, DCT and CD.

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Processes of Urine Formation:

Reabsorption of Important Substance:

- **Reabsorption of Glucose:**
  - Glucose is completely reabsorbed in the PCT.
  - It is transported by secondary active transport (**sodium cotransport**) mechanism
- **Reabsorption of Amino Acids:**
  - They are also reabsorbed completely in PCT.
  - They are reabsorbed actively by the secondary active transport mechanism along with sodium.

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Processes of Urine Formation:

Reabsorption of Important Substance:

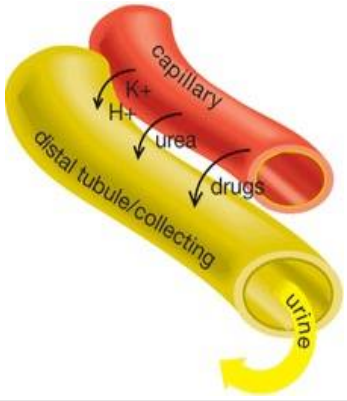
- **Reabsorption of Bicarbonate:**
  - Bicarbonate is reabsorbed actively, mostly in proximal tubule.
  - It is reabsorbed in the form of carbon dioxide.
  - Bicarbonate is mostly present as Na+ bicarbonate in the filtrate.

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Processes of Urine Formation:

3. Tubular secretion:

- The renal tubule and duct cells secrete other materials, such as wastes, drugs, and excess ions, into the fluid.



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Processes of Urine Formation:

3. Tubular secretion (tubular excretion):

- the process by which the substances are transported from **blood** into **renal tubules**.
- In addition to reabsorption from renal tubules, some substances are also secreted into the lumen from the peritubular capillaries through the tubular epithelial cells.

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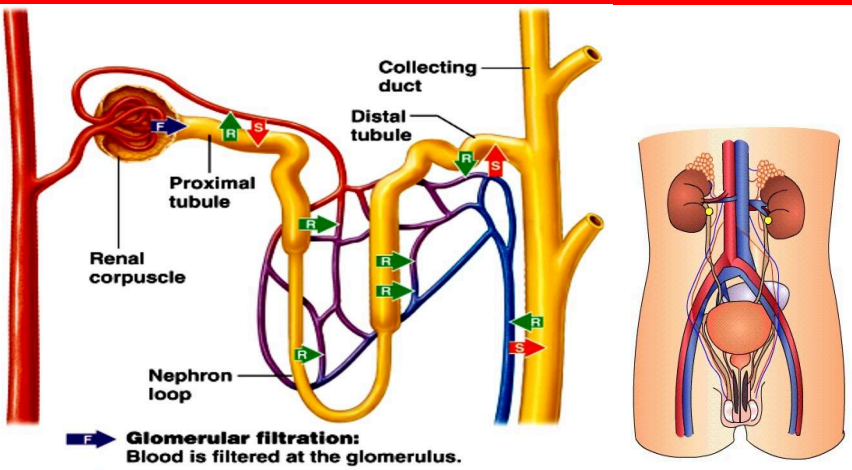
Processes of Urine Formation:

Substances Secreted in Different Segments of Renal Tubules

- **Potassium:** secreted actively by Na-K pump in PCT, DCT, and CD.
- **Ammonia:** secreted in the PCT.
- **H+ ions:** secreted in the PCT and DCT.
- **Urea:** secreted in loop of Henle.

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Urine Formation:



Glomerular filtration:
Blood is filtered at the glomerulus.

Tubular reabsorption:
Fluid and solutes are reabsorbed from the filtrate and returned to the blood.

Tubular secretion:
Substances are secreted from the blood into the filtrate.

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Questions/Comments



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